



AQUACULTURAL ENGINEERING

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH	IIT Kharagpur	8.46 / 10
2024	B.Tech	COAE,UAS Bangalore	7.568 / 10
2019	CLASS XII,CBSE	TVN,Varanasi	68.6%
2017	CLASS X,CBSE	TVN,Varanasi	7.8 / 10

PROJECTS

Experimental Characterization and Mathematical Modeling of Carngiform BCF Locomotion in Nile Tilapia | M.tech Thesis (Aug'26-Present)

- Developing a **computer-vision based workflow** to extract and reconstruct the body midline and caudal-fin motion of Nile tilapia from **60-fps top-view swimming videos**
- Quantifying **BCF swimming kinematics**, including tail-beat frequency, lateral amplitude, wavelength, phase propagation, wave speed, and body curvature
- Analyzed **body-wave propagation and swimming patterns** to characterize fish locomotion and develop a mathematical representation of swimming kinematics

INTERNSHIPS

i4-Marine Technologies Pvt. Ltd., Pune | Marine Technology Intern (May'26 -Jul'26)

- Designed and developed an **IoT-based Automatic Fish Feeder System** for automated feeding in aquaculture applications
- Designed **hopper and mechanical components using CAD**, integrating **ESP32/Raspberry Pi, Pixhawk, sensors, and motors** to develop an automated feeding system
- Performed hardware testing, system validation, and troubleshooting** to ensure reliable and consistent operation of the automated feeding system
- Developed practical understanding of **ROVs, autonomous marine systems**, and marine technologies through exposure to design, operation, and field applications

COURSEWORK INFORMATION

Fish Biology and Fish Culture Techniques | Open Channel Hydraulics and Coastal Engineering | Fish Biology and Fish Culture Lab | Water Quality management Practice | Principal of Fishing Technology | Zero Waste Food Processing Industry | Processing and Preservation of Aquacultural Products | Advance Aquaculture Technology | Planning and Design of Aquacultural Farms | Design of Aquacultural Facilities and Equipment | Food Handling and Packaging | Aquaculture Lab

SKILLS AND EXPERTISE

Languages: Python, SQL, MATLAB **ML/Data Science:** Scikit-Learn, Pandas, Numpy, Matplotlib, Seaborn
Software &Other: AnyLogic, Power BI, MS Excel, MS SQL, SolidWork, ANSYS, QGIS **IoT &Embedded System:** ESP32, Raspberry Pi

POSITIONS OF RESPONSIBILITY

General Secretary -Technology | Atal Bihari Vajpayee Hall of Residence,IIT-Kharagpur (2025-2026)

- Led technology initiatives and coordinated student Participation** across **500+ residents**, fostering teamwork and engagement in hall activities
- Organized and supported technical events and initiatives**, encouraging students to participate and collaborate in technology -driven activities
- Coordinated technical resources, student teams**, and event logistics to ensure effective execution of technology-driven activities

Teaching Assistant | Aquaculture Laboratory (sep' 26-Present)

- Conducted laboratory sessions and guided first-year PG student in **aquaculture experiments, and practical problem-solving**
- Supported hands-on training in **SolidWork, ANSYS, Python**, and **QGIS** for engineering analysis and visualization
- Assisted students in experimental setup, data collection**, analysis, and troubleshooting during laboratory work

AWARDS AND ACHIEVEMENTS

National AI/ML Hackathon 2026-Actively participated in the National AI/ML Hackathon, Jointly organized by the Ministry of panchayati Raj, NIC Geospatial Division, and IIT Triupati

CERTIFICATIONS

Basic of PLC (50 Hrs) | Siemens Centre of Excellence, GTTC, Bengaluru (Dec'22)
Internet of Things (40 Hrs) | Siemens Centre of Excellence, GTTC , Bengaluru (jan'23)

EXTRA CURRICULAR ACTIVITIES

- Volunteering at Smart India Hackathon (SIH) 2025 - Hardware Edition**, supporting the organization and smooth execution of the national-level hackthon
- Volunteering with NSS for a 7-days educational initiative**,teaching **visually impaired students** and supporting their learning and development
- Inter-College cricket Tournament winner**, Represent the college team and secured 1st position in an inter-college level cricket tournament
- Represent ABV at general championship IIT-Kharagpur in basketball, cricket, and football